Lecture 5

Introduction

What is Machine Learning (ML)?

[Mit97]

Definition

A computer program is said to learn from experience E[35] with respect to some class of tasks T, and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E.

[35] learn from experience = Belief Update

E: Data

- tabular data: $X \in \mathbb{R}^{n \times d}$ (also called 'structured' data)
- time series: $x \in \mathbb{R}^{T \times d}$ (e.g., stocks, motion)
  - $x[t]$: data at the time step t
- text: IMDB, WMT, SQuAD, ...
- image: $x \in \mathbb{R}^{3 \times H \times W}$ (3: RGB)
  - MNIST[36] ($1 \times 28 \times 28$, $n = 60k$, $C = 10$); FMNIST
  - CIFAR[37]-10 ($3 \times 28 \times 28$, $n = 60k$, $C = 10$)
  - CIFAR-100 ($3 \times 28 \times 28$, $n = 60k$, $C = 100$)
  - ImageNet ($3 \times 256 \times 256$, $n \sim 1.3M$, $C = 1000$)
  - ...
- video: $x \in \mathbb{R}^{T \times 3 \times H \times W}$
  - $x[t]$: image at the t-th frame
- graph: $x = (V, A)$
  - feature matrix: $V \in \mathbb{R}^{n \times d}$
  - adjacency matrix: $A \in \{0, 1\}^{n \times n}$
- speech, state, ..., multimodal data (e.g., image + text)

[36] Modified NIST (National Institute of Standards and Technology)
[37] Canadian Institute for Advanced Research

Preprocessing of Categorical Variables

- one-hot encoding (dummy encoding)

T: Task

T: to learn **representations/features/patterns**[38] from data

- classification, regression
- detection, segmentation
- translation, summary, image captioning
- image/video/text/speech generation
- policy
- ...

[38] ICLR = IC on Learning Representations

Classification and Regression

We want (a program) to make decisions or predict the outputs when the inputs are given.

Supervised Learning

- E: $S = \{(x_i{}^{39}, y_i)\}_{i=1}^n$
  - $x \in \mathcal{X}$: feature, covariate, or predictor
    - for the moment, think of it as $\mathbb{R}^d$[40]
    - $X = [x_1^\top; x_2^\top; \cdots; x_n^\top] \in \mathbb{R}^{n \times d}$: design matrix
  - $y \in \mathcal{Y}$: label, target, or response
  - classification: $\mathcal{Y} = \{1, 2, \cdots, C\}$[41]
  - regression: $\mathcal{Y} = \mathbb{R}$
- T: to learn a function h[42] from inputs $x \in \mathcal{X}$ to outputs $y \in \mathcal{Y}$
- P: population risk $L_{\mathcal{P}}(h)$
  - target function $f : \mathcal{X} \to \mathcal{Y}$
  - classification: $L_{\mathcal{P}}(h) = \mathbb{E}_{x \sim \mathcal{P}}[\mathbf{1}(h(x) \neq f(x))]$
  - regression: $L_{\mathcal{P}}(h) = \mathbb{E}_{x \sim \mathcal{P}}[(h(x) - f(x))^2]$[43]

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[39] Subscripts are used to refer to elements of a vector or members of a set.

[40] structured data or tabular data

[41] It is often $\{0, 1, \cdots, C-1\}$. For the binary case $C = 2$, we use $\{0, 1\}$ or $\{-1, 1\}$.

[42] h for hypothesis

[43] Uncertainty... MSE... Gauss distribution... MLE...

Classification + Uncertainty

- E: $S = \{(x_i\, y_i)\}_{i=1}^n$
  - $x \in \mathcal{X} = \mathbb{R}^d$
  - $y \in \mathcal{Y}$
  - classification: $\mathcal{Y} = \{1, 2, \cdots, C\}$
- T: to learn a conditional probability function $h(x) = [h_1(x), h_2(x) \cdots, h_C(x)]^\top$ given inputs $x \in \mathcal{X}$ where $h_c(x) := p(y = c \mid x)$
- P: population risk $L_{\mathcal{P}}(h)$
  - classification: $L_{\mathcal{P}}(h) = \mathbb{E}_{(x,y)\sim\mathcal{P}}[\ell(h(x), y)]$
  - cross-entropy loss: $\ell(h(x), y) = -\log h_y(x) = -\log p(y \mid x)$
    - negative log-likelihood (NLL)
    - MLE

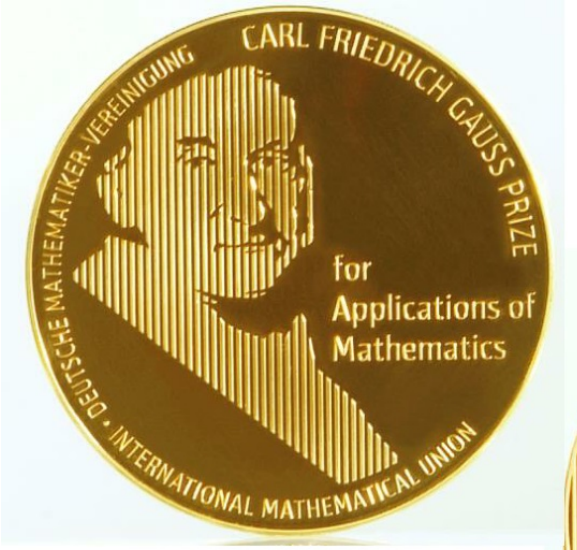
Regression + Uncertainty[45]

- E: $S = \{(x_i\, y_i)\}_{i=1}^n$
  - $x \in \mathcal{X} = \mathbb{R}^d$
  - $y \in \mathcal{Y}$
  - regression: $\mathcal{Y} = \mathbb{R}$
- T: to learn a function h[44] from inputs $x \in \mathcal{X}$ to outputs $y \in \mathcal{Y}$
- P: population risk $L_{\mathcal{P}}(h)$
  - target function $f : \mathcal{X} \to \mathcal{Y}$
  - regression: $L_{\mathcal{P}}(h) = \mathbb{E}_{x \sim \mathcal{P}}[(h(x) - f(x))^2]$ (MSE loss)
  - squared loss: $\ell(h(x), f(x)) = (h(x) - f(x))^2$
    - Gaussian noise: $y = f(x) + \varepsilon$
    - y: observation
    - $f(x)$: ideal
    - NLL/MLE

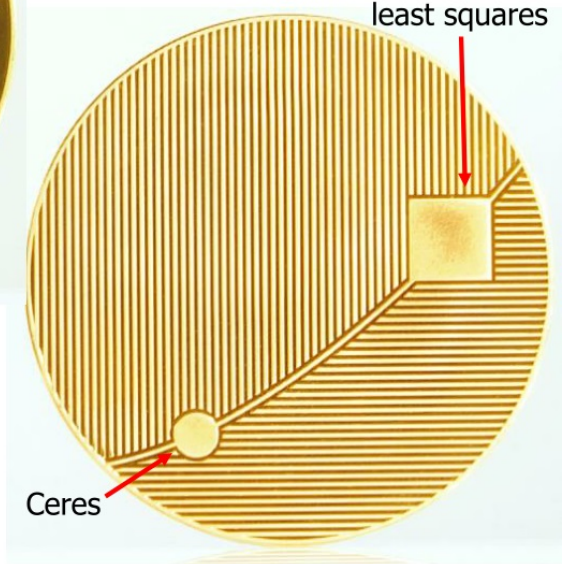
[44] h for hypothesis

[45] Continuous data has intrinsic stochasticity (e.g., ε). There are many other factors we don't know that determine the response.

The two loss functions, cross-entropy loss and MSE loss, are *the same* objective (NLL).



The Medal



Gauss Prize medal
designed by Jan Arnold

Figure: Left: Ceres, Right: Sketch of the orbits of Ceres, by Gauss.

- Giuseppe Piazzi (Jan. 1, 1801) discovered and tracked Ceres (19 observations) for 40 days before being lost in the glare of the Sun. He thought it was a new planet.
- The 24-years-old Carl Friedrich Gauss calculated the orbit of Ceres using Piazzi's observations, and Ceres was found near the predicted position on 31 Dec. 1801.

TABLE I

A Comparison between Maskelyne's Observations and Gauss's Calculations of the Minor Planet Ceres

1802		Right Ascension: Observation (Maskelyne)	Right Ascension: Calculation (Gauss)	Diff. (″)	Declination: Observation (Maskelyne)	Declination: Calculation (Gauss)	Diff. (″)
		° ′ ″	° ′ ″		° ′ ″	° ′ ″	
March	6	185 49 2·6	185 48 42·2	+20·4	16 3 49·5	16 3 56·1	−6·6
	14	184 15 3·8	184 15 35·1	−31·3	16 52 0	16 53 1·8	−61·8
	18	183 24 30·8	183 25 1·2	−30·4	17 14 10·6	17 14 15·2	−4·6
	25	181 54 18·8	181 54 39·0	−20·2	17 43 49·8	17 44 5·4	−15·6
April	6	179 28 22·8	179 29 8·2	−45·4	18 9 10·2	18 9 29·2	−19·0
	21	177 13 37·2	177 13 26·0	+11·2	17 51 41·4	17 52 8·9	−27·5
May	1	176 26 1·2	176 26 34·2	−33·0	17 12 34·7	17 12 26·6	+8.1
	4	176 19 5	176 19 29·7	−24·7	16 57 7·7	16 56 53·0	+14·7
	11	176 15 48	176 16 15·4	−27·4	16 12 41·8	16 15 4·2	−142·4
	13	176 18 6·9	176 18 45·8	−38·9	16 1 48·6	16 1 49·6	−1·0
June	20	181 2 7·8	181 1 15·7	+52·1	10 26 15	10 26 47·2	−32·2

Mean diff. in R.A. = 30″ Mean diff. in Decl. = 21″

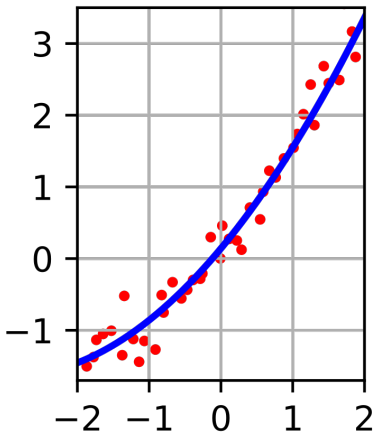


Figure: $30'' = \frac{1}{120}^{\circ}$

Least Squares: to calculate "orbit" using observations

Least Squares - MSE - Gaussian - MLE

$$
\begin{aligned}
y_i &= f_\theta(x_i) + \varepsilon_i \\
\varepsilon_i &\sim_{\text{i.i.d}} \mathcal{N}(0, \sigma^2) && \text{(Gaussian)} \\
\text{Lik}(\theta) &= \prod_{i=1}^{n} p(y_i \mid x_i; \theta) && \text{(MLE; iid-product)} \\
&= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(y_i - f_\theta(x_i))^2\right) \\
\text{NLL}(\theta) &= -\sum_{i=1}^{n} \log p(y_i \mid x_i; \theta) && \text{(NLL)} \\
&= \sum_{i=1}^{n} \frac{1}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}(y_i - f_\theta(x_i))^2 \\
&= \frac{n}{2}\log(2\pi\sigma^2) + \frac{n}{2\sigma^2}\text{MSE}(\theta) && \text{(MSE, Least Squares)}
\end{aligned}
$$

Least Squares - MSE - Gaussian - MLE - NLL - KL-divergence[46]

$$KL(p||q) := \mathbb{E}_p\left[\log \frac{p(Z)}{q(Z)}\right] \qquad \text{(KL divergence)}$$

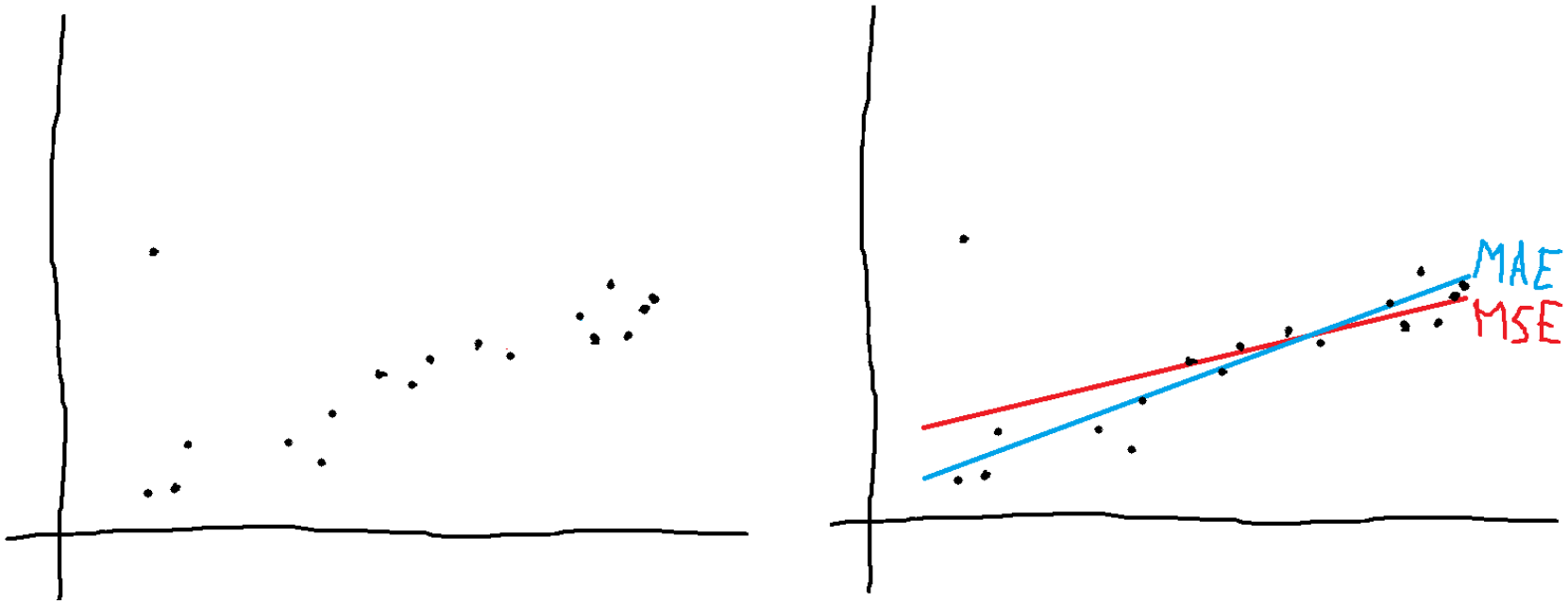
$$p_S(z) := \frac{1}{n}\sum_{i=1}^{n} \delta(z - z_n) \qquad \text{(empirical dist'n)}$$

$$\begin{aligned} KL(p_S||p_\theta) &= -H(p_S) - \mathbb{E}_{p_S}[\log p(Z;\theta)] \\ &= \underbrace{-H(p_S)}_{\text{constant}} \underbrace{-\frac{1}{n}\sum_{i=1}^{n} \log p(z_i;\theta)}_{\text{NLL}} \end{aligned}$$

[46] Ridge Regression - Weight Decay (ℓ_2-Regularization) - Gaussian (prior) - MAP - Constrained Optimization - Lagrange Multiplier

MSE loss is sensitive to outliers. We may need to use a different prior (assumptions on noise).

- ℓ_1 loss[47]
- Laplace distribution
- heavier (fatter) tail



[47] ℓ_1 : Laplace : Least Absolute Deviation : MAE : LASSO : median : sparse
$= \ell_2$: Gauss : Least Squares : MSE : ridge : mean : weight decay

How?

WANT: find a function h (T) with a good performance of $L_{\mathcal{P}}(h)$ (P)

BUT: we don't know the population distribution $\mathcal{P}$ and are only given a set of samples S (E).

- Use $L_S(h)$ instead of $L_{\mathcal{P}}(h)$
  - classification: $L_S(h) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}(h(x_i) \neq y_i)$
  - regression: $L_S(h) = \frac{1}{n}\sum_{i=1}^{n} (h(x_i) - f(x_i))^2$

Goal - Empirical Risk Minimization (ERM)

Find a minimizer $h_S \in \arg\min_h L_S(h)$

How can we find a minimizer over a function space?